

Seizure Prediction

ML Pipelines, Regularisation & Generalisation

3 EEG datasets · 2 preprocessing pipelines · Logistic regression baseline
L1 / L2 / Elastic Net regularisation · Class imbalance handling

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Agenda

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Dataset Collection

3 EEG datasets — size, imbalance, feature characteristics

02

Preprocessing Pipelines

Pipeline A (normalise-first) vs Pipeline B (extract-first)

03

Baseline Model

Logistic regression — accuracy, F1, PR-AUC, ROC-AUC

04

Overfitting & Underfitting

Regularisation sweep + learning curves

05

Regularisation Study

L1 vs L2 vs Elastic Net — sparsity & stability

06

Class Imbalance Handling

SMOTE, undersampling, class weighting — PR tradeoff

07

Comparative Analysis

Answering the four research questions

Dataset Collection

DS1

UCI Epileptic Seizure Recognition

Samples	11,500
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Features	178 (time-series)
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Class ratio	20% seizure
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Challenge	Moderate imbalance, dense signals
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DS2

EEG Brainwave Emotions (Kaggle)

Samples	2,132
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Features	2,548 (high-dim)
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Class ratio	33% minority
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Challenge	Extreme dimensionality vs sample count
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DS3

UCI EEG Eye State

Samples	14,980
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Features	14 (time-series)
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Class ratio	45:55 (balanced)
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Challenge	Near-balanced — generalisation test
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Preprocessing Pipelines

Pipeline A — Normalisation First

1

StandardScaler

Z-score normalisation across all features

2

Near-zero variance removal

Threshold = 0.01

3

SelectKBest (ANOVA F)

Retain top 30 features

Insight: Scaling before selection ensures fair feature ranking — prevents high-magnitude features from dominating.

Pipeline B — Feature Extraction First

1

Log-transform ($\log_{1p}$)

Positive shift before log

2

MinMax scaling

Normalise to $[0, 1]$

3

PCA (20 components)

Variance compression

Insight: Compressing variance before scaling retains semantically meaningful PCA components.

Baseline Model: Logistic Regression

+0.122

DS1 F1 Delta
Pipeline B vs A

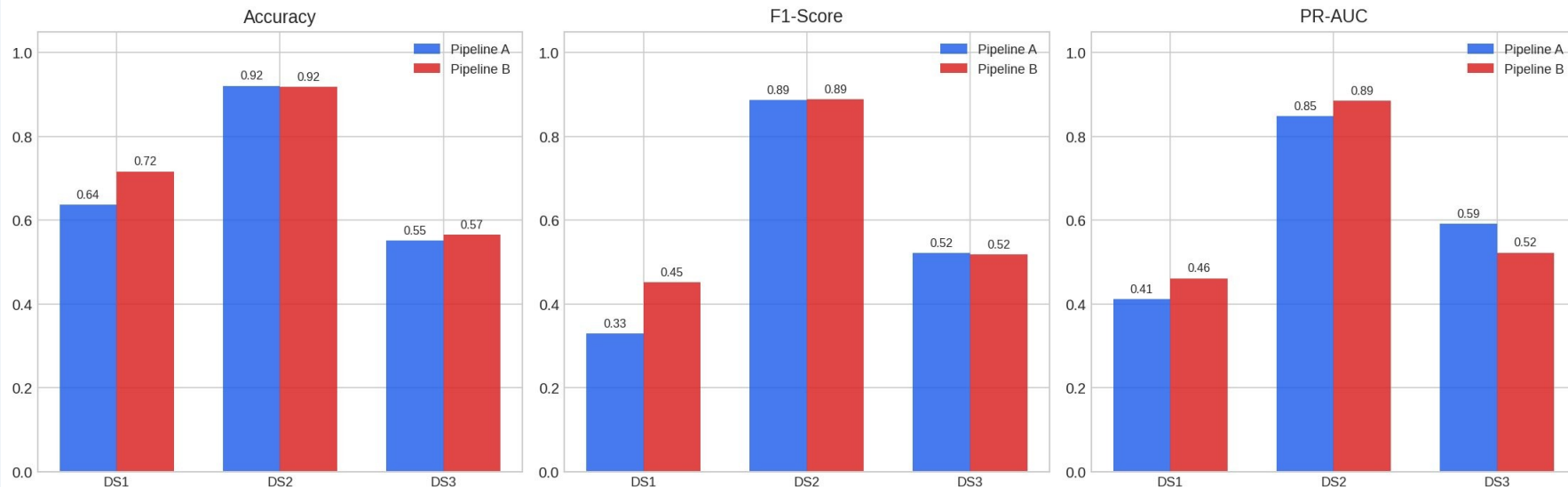
0.889

DS2 F1 Score
Both pipelines

0.591

DS3 PR-AUC
Pipeline A best

Figure 1: Baseline Logistic Regression — Pipeline A vs B



Overfitting & Underfitting Analysis

Underfitting Zone

$C=0.001$: train & test both ≈ 0.800

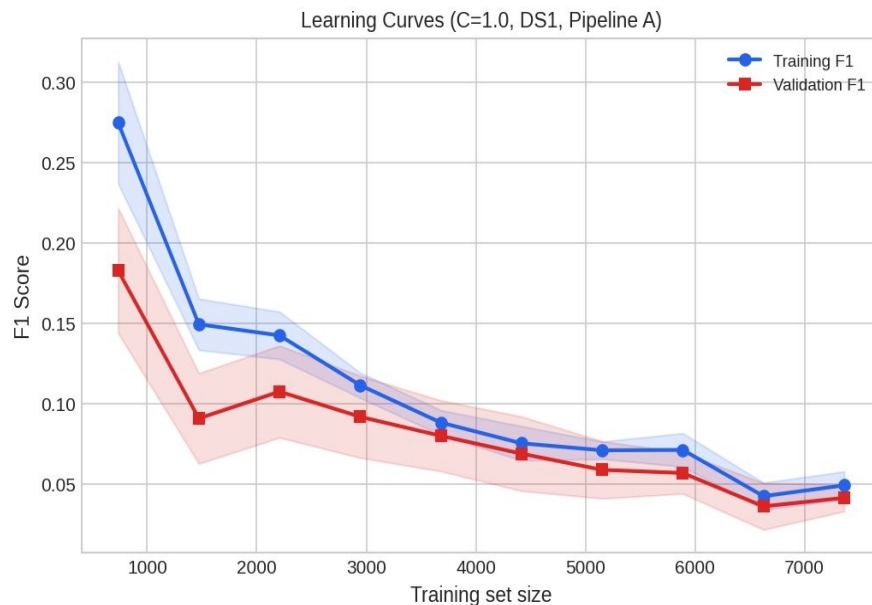
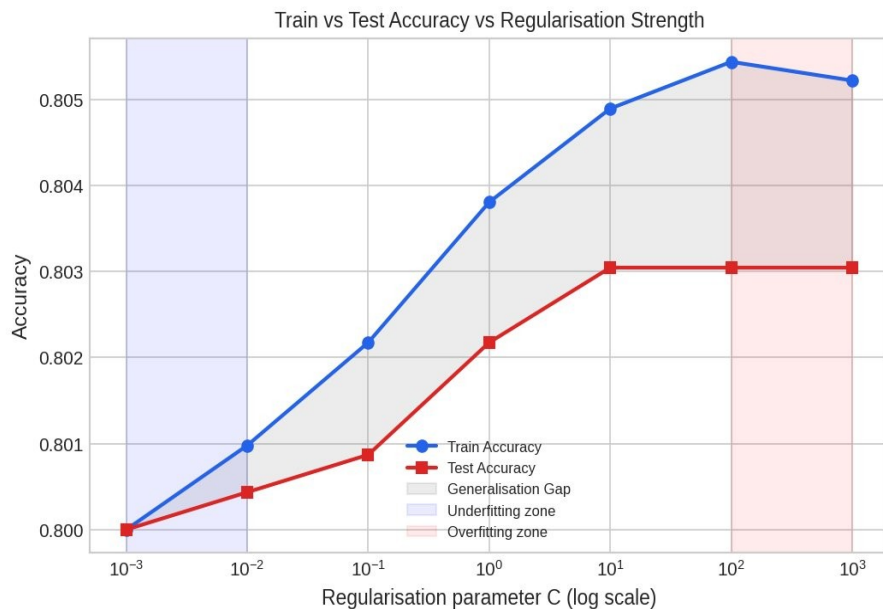
Overfitting Zone

$C \geq 100$: generalisation gap widens

Sweet Spot

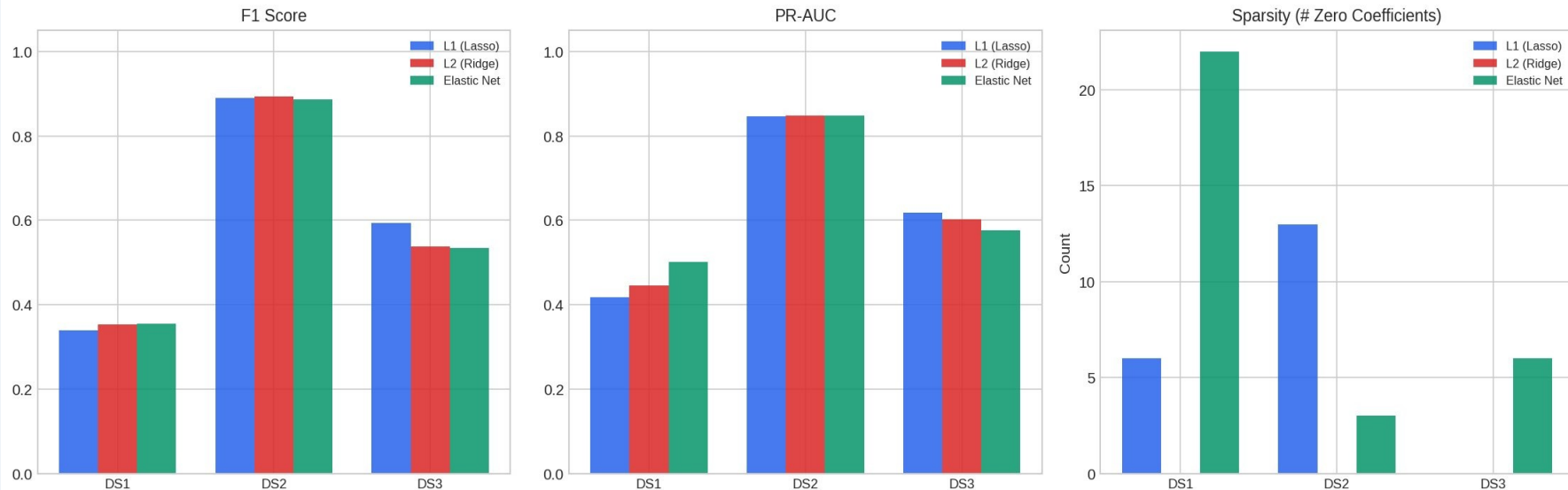
$C=10$: best test accuracy achieved

Figure 2: Overfitting & Underfitting Analysis



Regularisation Study: L1 vs L2 vs Elastic Net

Figure 3: L1 vs L2 vs Elastic Net — Cross-Dataset Comparison



DS1

Elastic Net

PR-AUC 0.502 — best on clinical data

DS2

L2 (Ridge)

F1 0.893 — handles high-dim well

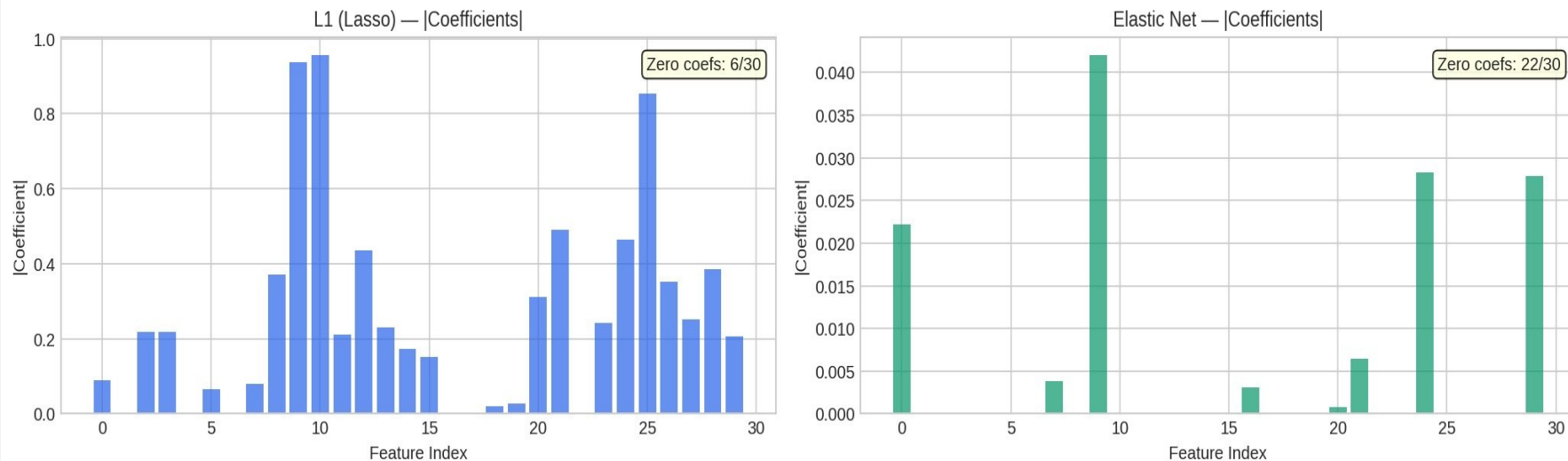
DS3

L1 (Lasso)

F1 0.593 — sparse features preferred

Coefficient Sparsity — L1 vs Elastic Net (DS1)

Figure 6: Coefficient Sparsity — L1 vs Elastic Net (DS1)



L1 (Lasso)

Zero coefs: 6/30

Moderate sparsity — retains broad feature coverage

Elastic Net

Zero coefs: 22/30

Aggressive selection — fewer, stronger features

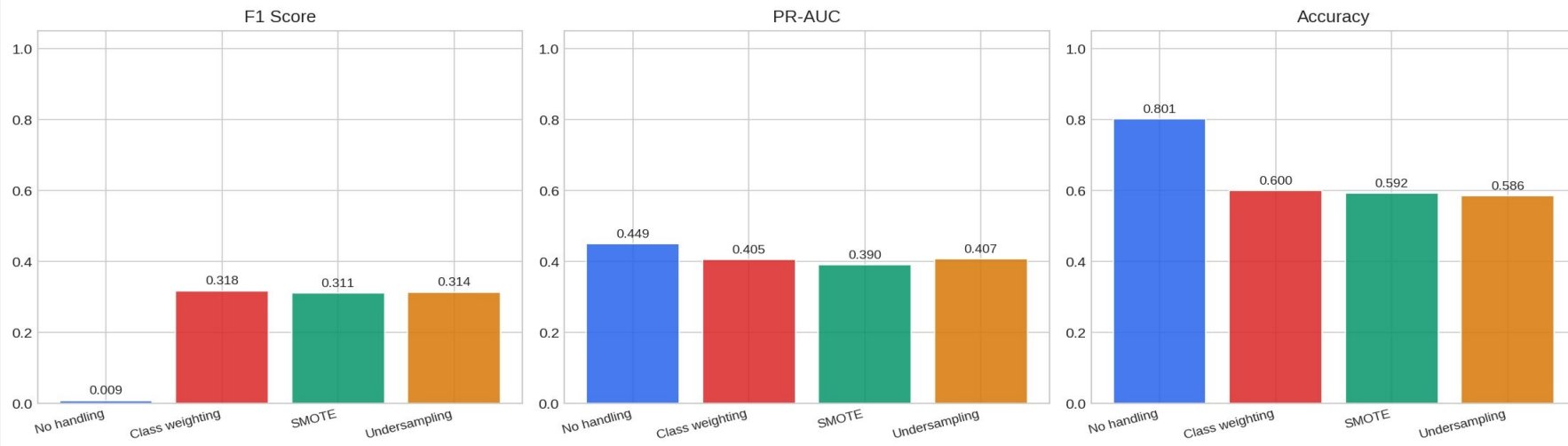
L2 (Ridge)

Zero coefs: 0/30

No sparsity — distributes weight across all features

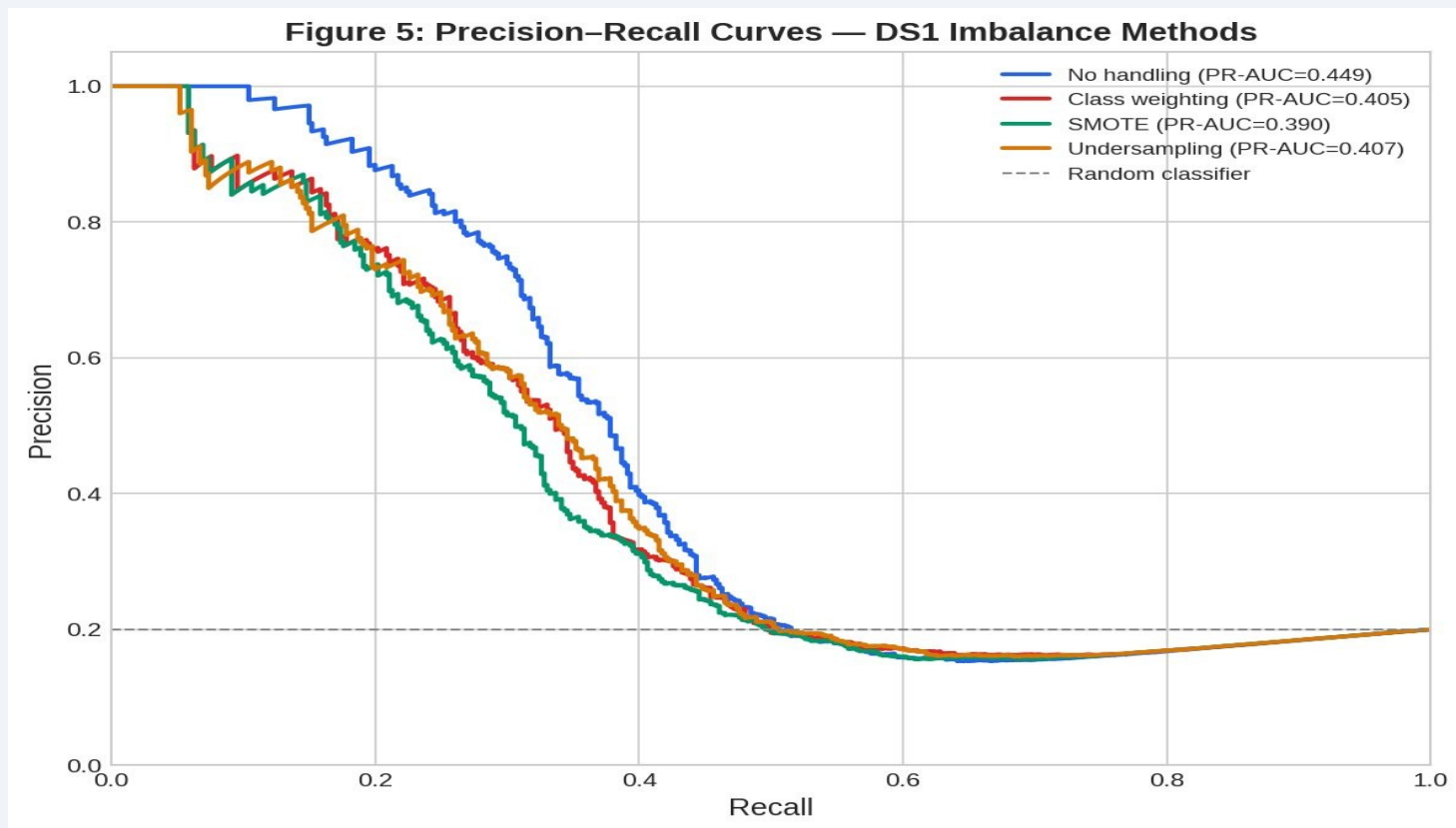
Handling Class Imbalance — DS1 (UCI Seizure)

Figure 4: Class Imbalance Handling — DS1 (UCI Seizure)



Method	Accuracy	F1	PR-AUC	Verdict
No handling	0.801	0.009	0.449	Predicts majority only
Class weighting	0.600	0.318	0.405	Best F1 — recommended
SMOTE	0.592	0.311	0.390	Comparable — slightly lower
Undersampling	0.586	0.314	0.407	Data loss risk

Precision-Recall Curves — DS1 Imbalance Methods



Note: No handling achieves highest PR-AUC (0.449) due to conservative precision at low recall — but F1 is near-zero. Class weighting is operationally superior.

Comparative Analysis — Research Questions

Q1: Does preprocessing order affect results?

YES — significantly for DS1. Pipeline B outperforms A by $F1 \Delta = 0.122$ (0.452 vs 0.329). Log-transform better handles EEG amplitude variance before PCA compression. Effect is negligible on DS2/DS3 ($\Delta < 0.005$).

Order matters most on high-variance, skewed data.

Q2: Which regularisation generalises best?

No single winner across all datasets. L1 wins DS3 ($F1=0.593$), L2 wins DS2 ($F1=0.893$), Elastic Net wins DS1 ($PR-AUC=0.502$). Selection should be dataset-specific via cross-validation.

Elastic Net is best on the clinical seizure dataset.

Q3: Does Elastic Net consistently outperform L1/L2?

NO. Elastic Net excels when features are correlated + sparse signal exists simultaneously (DS1 conditions). On DS2 (dense regularisation suffices) and DS3 (small, uncorrelated feature space) it does not win.

Benefit is conditional on feature correlation structure.

Q4: How does imbalance handling interact with regularisation?

Class weighting + Elastic Net is the most robust combo. L1-sparse models benefit more from SMOTE (clean class boundaries). Undersampling loses majority-class info and performs worst in data efficiency.

Class weighting + Elastic Net = recommended clinical strategy.

Conclusion

Preprocessing order: Large effect on raw time-series (DS1 F1 $\Delta=0.122$); minimal on pre-engineered features

Best regulariser: No universal winner — Elastic Net best on clinical DS1 (PR-AUC 0.502)

Elastic Net consistency: Not consistent — benefit conditional on feature correlation & sparsity

Imbalance correction: Class weighting most practical — best F1 without data augmentation or loss

Future work: Non-linear models (RF, SVM-RBF) and raw EEG feature engineering